

2.3.1 Decomposition

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔

CS XI

Chapter 2: Computational Thinking & Algorithm

Lecture 04 of 13

★ Golden

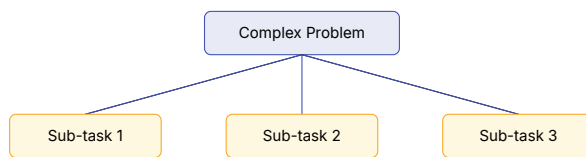
2.3.1 Decomposition ★ Golden

2.3 Algorithmic Strategies for Problem Solving. An **algorithmic strategy** is a general method that helps you design an algorithm. The three strategies in your syllabus are decomposition, pattern recognition and abstraction.

LEARN IT

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts. By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve.

Each small part is called a **sub-task** (or sub-problem). You solve each sub-task separately, then join the solutions to solve the whole problem.



Decomposition: one complex problem is split into smaller sub-tasks.

WORKED EXAMPLE

Real-life example: Learning a new language

"Learn English" (or any language) is too big to do in one go. Decompose it:

Sub-task	Description
Learn vocabulary	Learn the words for people, actions and things (e.g., "I", "eat", "apple").
Grammar rules	Learn the sentence structure: Subject → Verb → Object.
Learn tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence building	Combine words using the rules to create valid sentences: "I ate an apple."

Vocabulary → Grammar rules → Tenses → Sentence building

WORKED EXAMPLE

Computing example: a "Student Result" program

Problem: read the marks of a student in 5 subjects and print the total, percentage and grade.

- Sub-task 1: Input marks of 5 subjects.
- Sub-task 2: Calculate total = sum of the 5 marks.
- Sub-task 3: Calculate percentage = $\text{total} \div 500 \times 100$.
- Sub-task 4: Decide the grade from the percentage.
- Sub-task 5: Print the result card.

Each sub-task is now so small that you can write it in two or three lines of code.

Why is decomposition useful?

- Reduces complexity:** you focus on one piece at a time.
- Helps in debugging:** you can find exactly where the mistake is (e.g., in the language example, is the error in the verb tense or the word order?).
- Promotes reusability:** once you learn or build a small part, you can apply it elsewhere (the "calculate percentage" part can be reused in an attendance program).

REMEMBER IT

Think of eating a big *roti*: you cannot eat it in one bite, so you break it into small pieces. Decomposition = "break the roti".

COMMON MISTAKE

Do not break a problem into random pieces. Each sub-task must be a meaningful, clear job, and all sub-tasks together must solve the *whole* problem. If one part is missing (e.g., "print result"), the solution is incomplete.

اردو میں سمجھیں

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔ مثلاً نئی زبان سیکھنے کو الفاظ، گرامر، زمانے (Tenses) اور جملہ سازی میں تقسیم کیا جا سکتا ہے۔ اس سے پیچیدگی کم ہوتی ہے، غلطی ڈھونڈنا آسان ہوتا ہے اور حصے دوبارہ استعمال ہو سکتے ہیں۔

EXAM TIP

Expect "Define decomposition with a real-life example" or "Explain the benefits of decomposition". Give the definition, draw the small tree (Complex Problem → Sub-task 1, 2, 3), show the language-learning table, and list the three benefits: reduces complexity, helps debugging, promotes reusability.

CHECK YOURSELF

- What is each smaller part of a decomposed problem called?
- Decompose "Making a school timetable" into three sub-tasks.
- State the three benefits of decomposition.

Answers: 1. A sub-task (sub-problem). 2. Example: list subjects and teachers; decide periods per subject per week; place periods in days so no teacher has two classes at once. 3. Reduces complexity, helps in debugging, promotes reusability.

Next lecture: 2.3.2 Pattern Recognition